

Examining Wealth Clustering in Tanzania

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1 Introduction

Given a wealth of demographic and survey data in the developing world, much research in economics and international development focuses on identifying the causes and correlates of wealth, poverty, and income (Newhouse, 2024). Of particular interest is to estimate household wealth for sub-national geographic areas to better target foreign development aid through organizations like the World Bank and International Monetary Fund (IMF) (Masaki et al., 2022; Newhouse, 2024). In this project, we continue this tradition by studying the geographic clustering of household wealth using survey data from Tanzania. We investigate two main research questions: 1) To what extent is household wealth clustered within enumeration areas (EA) and 2) How much of the clustering can be explained by additional household characteristics, such as bank account possession, or educational attainment? Understanding the extent to which wealth is clustered across these EAs is essential to inform the design and targeting of development policies. Empirical evidence of wealth clustering within EAs would inform government officials in targeting policy interventions to address inequalities. If we find no evidence of wealth clustering within these EAs, then wealth disparities may be driven by household-level factors, suggesting that policymakers focus on improving decision-making around resource use and distributions at the household level.

2 Data Set

2.1 Data Description

We analyzed household data from the 2022 Tanzania DHS, which randomly selected 629 EAs from all the 31 regions across the mainland and Zanzibar [A.1], completing interviews in 15,705 households. Many variables containing information about household amenities and facilities and individuals in the household are included along with sampling weights. The DHS provides detailed documentation on each variable.

2.2 Variable Selection and Exploration

Our outcome variable is the Wealth Index, a composite measure of a household’s cumulative living standard, an often used measure of socioeconomic standing in the international development literature (Rutstein and Johnson, 2004). It is calculated using data on a household’s ownership of selected assets and dwelling characteristics related to wealth status. Each asset is assigned a weight or factor score generated through principal components analysis which are then standardized. Each household is assigned a standardized score for each asset, differing depending on whether or not the household owns the asset, which are summed by household and individuals ranked according to their total household score. We also consider household-level covariates, including the sex and educational attainment of the household head; number of members; and household possession of a bank account. As most dwelling characteristics were used to calculate the wealth index, we exclude them from our covariate set. We also compute a dependency ratio feature for each household, a demographic measure that compares how many people in the household are dependents (ages 0 to 14 and over 65) relative to the expected working-age caretakers (ages 15 to 64) (Shryock and Siegel, 1976). The ratio between the counts is multiplied by 100 and capped at 100 to resolve divisions by 0 for houses with no recorded working-age individuals. Capping the variable also controls it, as it is usually computed at an aggregate level and not the household level. The original maximum ratio was 900, and this reduction affected 5,038 households. We selected three separate indicators for residence which are described and summarized in Table 1. Only one "ADMIN" level is shown for brevity, but all were represented by at least 418 households [Table A.1].

Data exploration revealed differences in wealth distribution between administrative regions [A.2], urban versus rural strata and analytic regions, education of the household head [A.3] and bank account possession [A.4]. Higher dependency ratios appeared to correspond to lower average wealth when aggregating by region [A.5]. Aggregation also revealed a slight positive relationship between household size and wealth. Discrepancies in wealth were less evident between the sexes [A.6]. Multivariate plots implied stronger predictive power for bank account possession than for household size [A.7]. The wealth

Table 1: Variable Descriptions and Summary Statistics

<i>Panel A. Continuous Variables</i>				
Variable	Description	Mean	Domain	Std. Dev
HHWEALTH	Standardized Wealth Index	0	[-174173,354602]	99713.6
HHSIZE	Household size (number of members)	4.69	[1,54]	2.848
DEPRATIO	Dependency ratio, ages 15-64 to 0-14 and 65+	69.425	[0,100]	38.903
<i>Panel B. Categorical Variables</i>				
Variable	Description	Level	Proportion	Count
HHEAD_SEX	Sex of household head	Male	0.712	11,183
		Female	0.288	4,523
HHEAD_EDUC	Educational attainment of household head	None/Early	0.196	3,081
		Primary	0.577	9,068
		Secondary	0.193	3,027
		Higher	0.25	394
		Don't know	0.09	135
HASBANK	Household has bank account	No	0.803	12,611
		Yes	0.197	3,094
RES_TYPE	Urban or rural residence	Urban	0.329	5,171
		Rural	0.671	12,534
ADMIN	Which of the 31 administrative regions	Dar es Salaam	.052	828
		...	...	...
ANA_REG	Which of the 4 analytical regions	Mainland Rural	0.570	8,957
		Mainland Urban	0.286	4,493
		Pemba Island	0.055	874
		Zanzibar Island	0.089	1,381

index itself was not normally distributed, exhibiting a considerable right skew toward poorer households (Figure 1). There were *no missing values* for any variable.

3 Modeling Approach

We are interested in how much wealth variability exists *between* EAs, so we implemented a multilevel modeling approach that models random intercepts for each EA. We can therefore examine how much between-EA variability is explainable through the random effects, thus capturing the effects of geographic clustering on household wealth. We then augment the models with additional household and geographic contextual variables to examine whether we can explain variation in household wealth related to geographic location in excess of household characteristics.

To introduce notation, let $\{i, j\}$ denote a household i in EA j . We model a household's wealth $Y_{i,j}$ using a multilevel framework. Each model included fixed effects for bank account, sex and education of household head, number of members, dependency ratio, and random intercepts by EA. The “number of members” variable was centered at 0 and wealth index was divided by 100,000 for to simplify interpretations. $\mathbf{X}_{i,j} =$

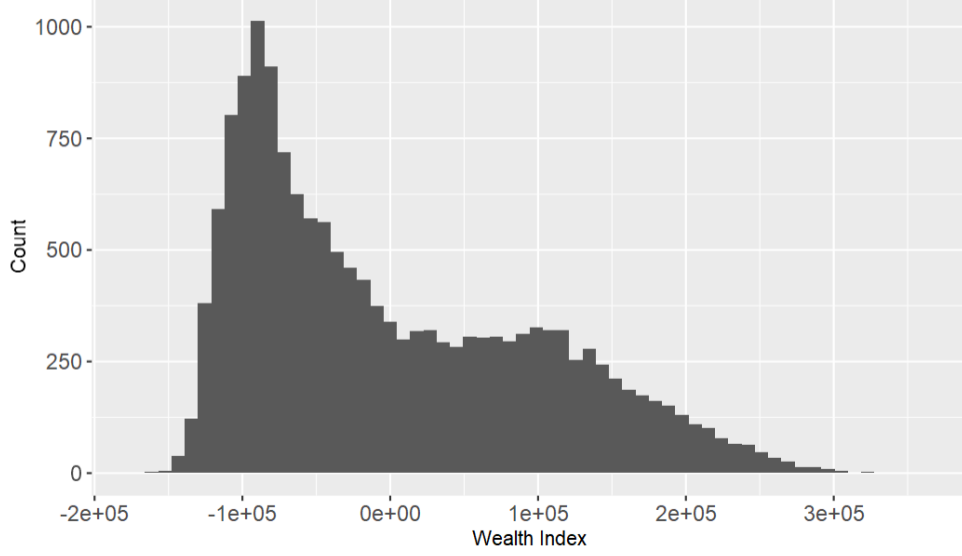


Figure 1: Distribution of Wealth Score

$[X_{1,i,j}, \dots, X_{K,i,j}]$ is a vector of K household-level characteristics, including demographics and geography as shown in Table 1. For each model, $b_{oj} \sim N(0, \sigma_0^2)$ and $e_{ij} \sim N(0, \sigma^2)$, with b_{oj} denoting random intercepts and e_{ij} denoting random error.

- **Model 1:** Includes fixed effects for administrative region. The variability between EAs is especially important within these regions in terms of policy evaluation, as significant variability in household wealth *within* a specific region helps policymakers determine which localities within them to target. Note that $\beta = [\beta_1, \dots, \beta_{38}]$.

$$Y_{ij} = \beta_0 + \mathbf{X}_{ij}\beta + b_{oj} + e_{ij}, \quad (1)$$

- **Model 2:** To reduce the number of effects being estimated, "ANA_REG" replaced "ADMIN" from Model 1. Notation is the same as Model 1, but now $\beta = (\beta_1, \dots, \beta_{11})$ to reflect the restricted regions.
- **Model 3:** Nearly identical to Model 1, but included a random effect for bank account possession, and a fixed effect for urban/rural "place of residence". Motivated by the fact that certain (richer) regions typically have a much higher rate of households with bank accounts. The random slope variance and covariance between random effects are denoted by σ_1^2 and σ_{01} , respectively. Here, $\beta = [\beta_1, \dots, \beta_{39}]$.

$$Y_{ij} = \beta_0 + \mathbf{X}_{ij}\beta + b_{oj} + b_{1j}\text{HAS_BANK}_{ij} + e_{ij} \quad (2)$$

We are primarily interested in the variance components σ_0^2 and σ^2 through which we can estimate how much variability between EAs pertains to geographic clustering, net of the covariates. Further, we omit the DHS sampling weights from our analyses for two reasons. First, we are interested in examining household-level correlates of wealth and not nationally-representative of Tanzania as a whole. Second, we found that including fixed effects related to the DHS sampling design (region and urban/rural residence) yielded nearly identical weighted and unweighted results. We therefore opt for a model-based inferential approach and estimate unweighted multilevel models.

As our primary interest is to measure geographic clustering in household wealth, we first examine the appropriateness of our random effects specifications. This involves likelihood ratio tests of the null hypothesis that the random intercepts can be constrained at zero.¹ We first test two baseline models without covariates. The first model is a baseline specification that includes only district-level fixed effects while the second adds in EA-level random intercepts. In line with our hypothesized geographic clustering, we find a significant likelihood ratio test that rejects the null hypothesis that the EA-level random intercepts are zero. This is true for both the baseline model and after including covariates. We thus conclude that geographic clustering in Tanzanian household wealth is a statistically significant factor, even after accounting for household contextual variables.

4 Interpretation and Inference

Table 2 provides a summary of the relevant parameter estimates and standard errors for each model. We omit administrative and analytical region fixed effects from the table for brevity, as they are primarily nuisance parameters. Most of these were insignificant or trivial, based on urban or rural distinction.

4.1 Empirical Findings

We first examine the model covariates, finding that nearly all symptomatic indicators were statistically significant predictors of a household's wealth, net of region-specific effects. The interpretation of the overall intercept changes between models as each used

¹Because the null hypothesis lies on a boundary of the random effect variance σ_0^2 parameter space, we use a mixture χ^2 distribution.

Table 2: Multilevel Regression Estimates

Variable	Model 1		Model 2		Model 3	
	Estimate	Std. Err	Estimate	Std. Err	Estimate	Std. Err
Intercept	-0.3736	0.1197	0.4196	0.0332	0.3442	0.0827
Has Bank Acct.	0.7102	0.0117	0.7004	0.0117	0.7095	0.0175
Head Female	-0.0089	0.0093	-0.0099	0.0093	-0.0098	0.0092
Num. Members (centered)	0.0159	0.0016	0.0159	0.0017	0.0159	0.0017
Dependency Ratio	-0.0008	0.0001	-0.0008	0.0001	-0.0007	0.0001
Head Education:						
Primary	0.1557	0.0113	0.1587	0.0112	0.1579	0.0111
Secondary	0.4141	0.0149	0.4180	0.0149	0.4014	0.0147
Higher	0.6816	0.0299	0.6898	0.0298	0.6561	0.0303
Rural Household	-	-	-	-	-0.9449	0.0335
σ^2	0.2368		0.2368		0.2265	
σ_0^2	0.3020		0.1561		0.1217	
σ_1^2	-		-		0.0826	
σ_{01}	-		-		-0.0030	
ICC	0.560		0.397		NA	
AIC	24318.53		23841.92		23575.68	
BIC	24631.66		23949.18		23929.01	

different reference categories.

For Model 1, we find that the averaged-sized, male-headed household in the Dodoma (capital) Region with no dependents, no bank account, and none/early education falls -0.374 standard deviations below Tanzania’s average wealth. The intercept for Model 2 has a similar interpretation, but the baseline for the geographic variable is now the urban mainland, changing the estimate to a positive value of 0.4196 standard deviations. Model 3 uses the same reference categories as Model 1, but adds the stipulation of an *urban* home within Dodoma through the urban/rural indicator, explaining the positive parameter estimate. The sign change in intercepts highlights the importance of urban versus rural residence in wealth discrepancies (Carletto et al., 2022).

Other covariates interpretations are similar across all models in direction and magnitude. Most demographic predictors were statistically significant, consistent with extensive development economics literature (Carletto et al., 2022). In each specification, we find that wealth was between 0.70 - 0.71 standard deviations higher for households with a bank account, holding other variables constant. For each model, presence of a bank account was the strongest predictor of wealth. Households with a female household head had a negative but small coefficient estimate; however, with a small p -value we fail to reject the null hypothesis that this difference is nonzero. Moving to number of members,

we find that each additional household member above the national average was associated with a 0.016 standard deviation higher average wealth. Comparatively, a large dependency ratio, all else equal, was negatively associated with household wealth, suggesting that households with more dependents relative to providers lower average wealth. Educational attainment of the household head was also highly predictive of household wealth and showed consistent effect sizes across model specifications. Average wealth in households with a high (tertiary) level of education was 0.65-0.68 standard deviations higher than the Tanzanian average. Indeed, even lower levels of educational attainment showed higher average wealth of around 0.16 standard deviations for primary education and 0.40-0.42 for secondary education, holding other factors constant. Residual variance was nearly identical across all models, but smallest in Model 3. These values represent the leftover variability in wealth index not explained by the parameters in the models, from which we can estimate how much remaining residual variability can be explained by geography.

4.2 Identifying Geographic Clustering

To see how household wealth is clustered between households within districts, we examine how the EA-level random effects explain variability in household wealth on the *intensive margin*. That is, how much mean household wealth differs between households within each district. Specifically, we compare how much of the total residual variability in our models can be explained by the random intercepts using intra-class correlation coefficients (ICCs). These represent the share of residual variance not explained by the district fixed effects or the relevant covariates. The remaining variability explained by the EA random effects thus represents the extent to which geographic clustering explains the remaining variation in household wealth. Note that ICCs are not appropriate metrics for models with random slopes. Each model had a notable level of variability in the intercepts, indicating wealth clustering in Tanzania where some EAs have considerably higher or lower wealth than others. Lower levels of variability for Models 2 and 3 are attributable to the analytic regional variable in the former and a rural indicator in the latter, both of which capture less variability than the full set of administrative regions.

Further, the rural indicator variable deflates the predictive power of other covariates. For Model 3, the variance in slopes is relatively small, suggesting the effect of bank account possession is largely the same across EAs, though its inclusion reduced residual variance compared to the other models. The covariance between slopes and intercepts is very small, meaning the baseline wealth index of a household is not related to the strength of effect for bank account.

4.3 Model Validation

Model 3 fits the data the best in terms of AIC and BIC. Lower ICC values represent a greater contribution of the random effects explaining variability. An ICC of 0.377 for Model 2 means that over 60% of the remaining availability is explained by the random effects, a greater extent to which geographic clustering explains the remaining variation in household wealth than Model 2. This suggests that the administrative region fixed effect is doing much of the work that the random intercepts would be doing for Model 1. The analytical region fixed effect in Model 2 captures less variability in wealth, so the random intercepts have to compensate. To validate the results of the models, we evaluated standard assumptions on the random effects and residuals. According to the Quantile-Quantile plots (Figure 2), the residuals of the model have heavy tails but generally appear to follow a normal distribution, which is corroborated by histograms [A.8]. The scatter plots show a slight downward trend between residuals and fitted values - there are larger residuals for smaller fitted values (lower wealth). There is also a large cluster of points around 0 residual and -1 scaled wealth index. These trouble the assumption of homoscedasticity because variance is not constant across residuals. This is likely a result of the right skew of the wealth index, where wealth is more concentrated toward poorer individuals. This combined with several wealthy outliers in the sample leads to the behavior seen in the scatter plot. Future procedures would need to either normalize wealth index or use more flexible generalized linear models that are appropriate for fitting non-normal outcomes. Models 2 and 3 had similar residual diagnostics [A.9].

The intercepts for Models 2 and 3 seem to have the “best” normality, with theoretical quantiles aligning most closely with the sample quantiles (Figure 3). The intercepts for

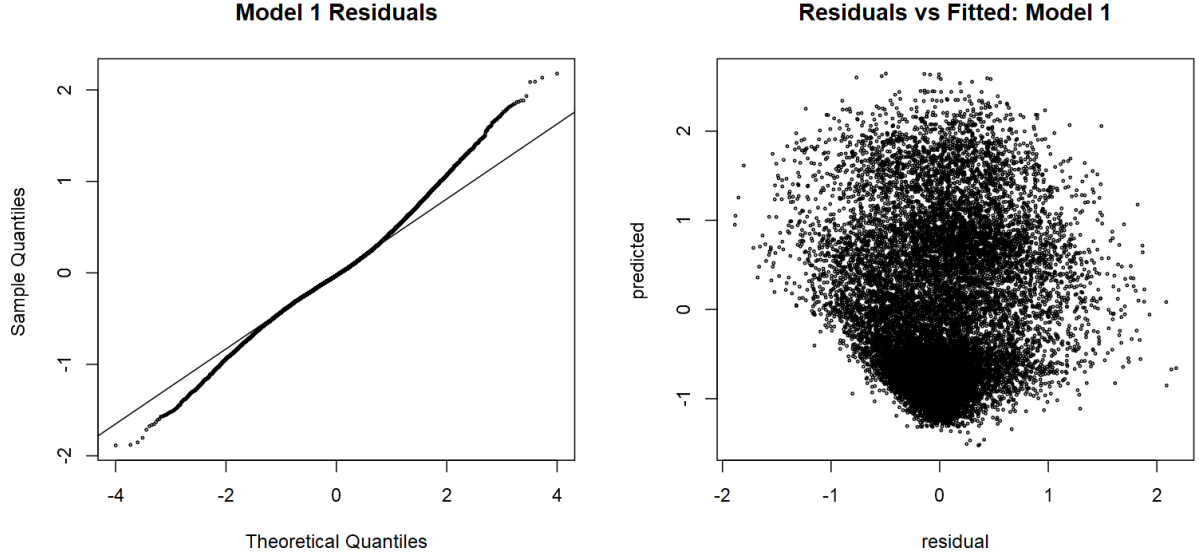


Figure 2: Model 1 check for Residual Normality, Homoscedasticity

Model 1 and slopes for Model 3 seem to have extremely heavy tails and deviations around the center of the distribution, suggesting that Model 2 has better residual diagnostics and thus more reliable fit results. A closer inspection of the Model 1 intercepts reveals bimodality [A.10], likely a result of urban/rural wealth discrepancies. The peak is higher for lower values because there are more rural (poorer) households in the sample than urban (richer) households. The considerable difference in intercepts is why the distribution appears as a sort of "mixture". Models 2 and 3 do not have this issue because the fixed effects for analytic region (Model 2) and residence type (Model 3) capture urban/rural differences more efficiently than administrative region for Model 1.

5 Conclusions

Through our multilevel modeling approach we found clear evidence for wealth clustering in Tanzania. We found that after controlling for common household and demographic variables, random intercepts for each EA accounted for substantial residual variance—a clear sign of geographic clustering. This was especially evident in Model 2, in which around 60% of the residual variance was attributed to geography. Indeed, the corresponding ICC was 0.397, suggesting that household wealth is highly clustered after accounting for both household demographic factors. Our conclusions were robust to a few model specifications that fit the data well.

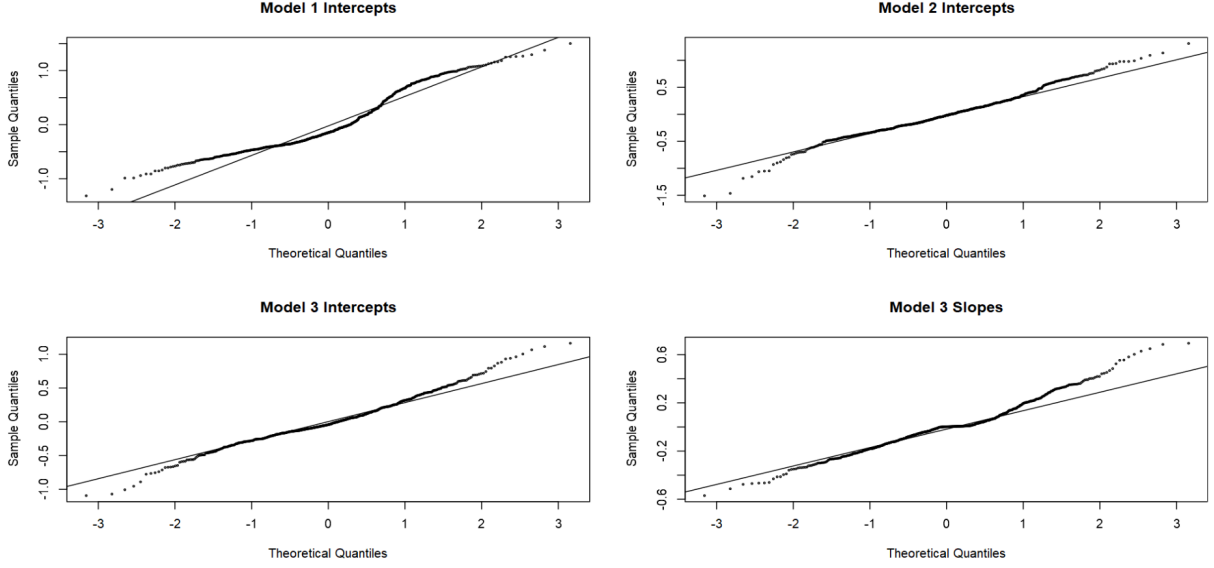


Figure 3: Normality of Random Effects

Having identified geographic clustering, we also examined various household and geographic features associated with household wealth. Among our set of explanatory variables, we identified possession of a bank account, educational attainment, residence in an urban areas, dependency ratio, and, to a lesser extent, number of members as key predictors of household wealth. Since most of these indicators reflect a household’s socioeconomic standing ([Rutstein and Johnson, 2004](#)), it was unsurprising to find them as significant predictors of household wealth. We also found slight heterogeneity in the effect of bank account possession related to a household’s location. Including random slopes that vary by EA showed that we could explain additional residual variation in wealth. This suggests that improving financial access to households in Tanzania may have a differing impact depending on location. Policymakers and multilateral organizations may therefore find a bigger “bang for their buck“ by taking geographic clustering into account when targeting interventions to improve financial access in the developing world. Our results also emphasize the value of investment in education to improve wealth, and suggest that wealth is not significantly determined by sex of the household head.

The primary limitation of our project is the exclusion of sampling weights, preventing us from making inference for the entire country. While future work may include sampling design features, our findings provide a useful starting point for further analysis.

References

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- Masaki, T., Newhouse, D., Silwal, A. R., Bedada, A., and Engstrom, R. (2022). Small area estimation of non-monetary poverty with geospatial data. *Statistical Journal of the IAOS*, 38(3):1035–1051.
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A Figures and Tables

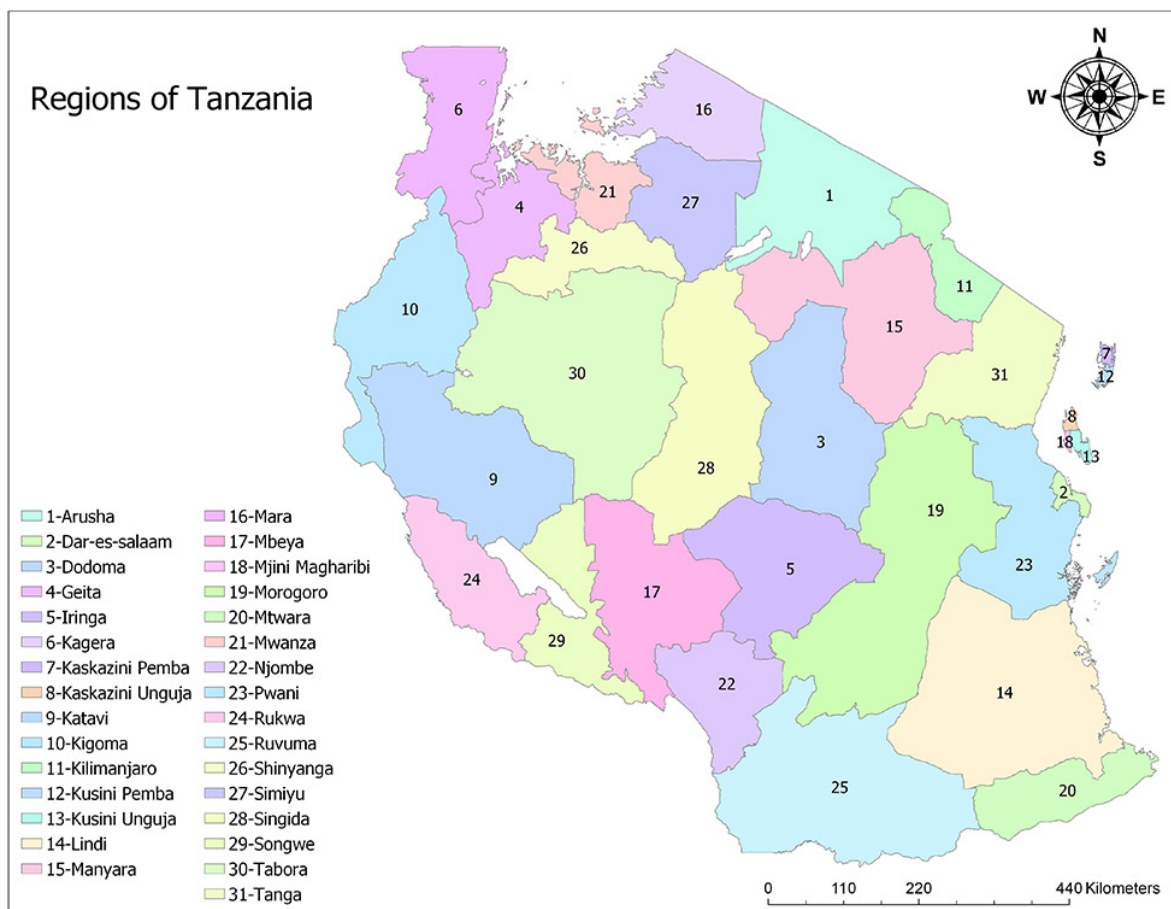


Figure A.1: Tanzanian Regions

Table A.1: Administrative Regions sample proportions

ADMIN	Region	Proportion	Count
	Arusha	0.0315	495
	Dar es Salaam	0.0527	828
	Dodoma	0.0356	560
	Geita	0.0313	492
	Iringa	0.0282	443
	Kagera	0.0363	571
	Kaskazini Pemba	0.0267	420
	Kaskazini Unguja	0.0291	457
	Katavi	0.0294	463
	Kigoma	0.0320	503
	Kilimanjaro	0.0315	495
	Kusini Pemba	0.0289	454
	Kusini Unguja	0.0266	418
	Lindi	0.0305	480
	Manyara	0.0326	512
	Mara	0.0308	484
	Mbeya	0.0329	517
	Mjini Magharibi	0.0322	506
	Morogoro	0.0353	555
	Mtwara	0.0344	541
	Mwanza	0.0347	546
	Njombe	0.0305	480
	Pwani	0.0311	489
	Rukwa	0.0297	466
	Ruvuma	0.0321	508
	Shinyanga	0.0326	513
	Simiyu	0.0309	486
	Singida	0.0305	480
	Songwe	0.0295	464
	Tabora	0.0331	520
	Tanga	0.0356	559

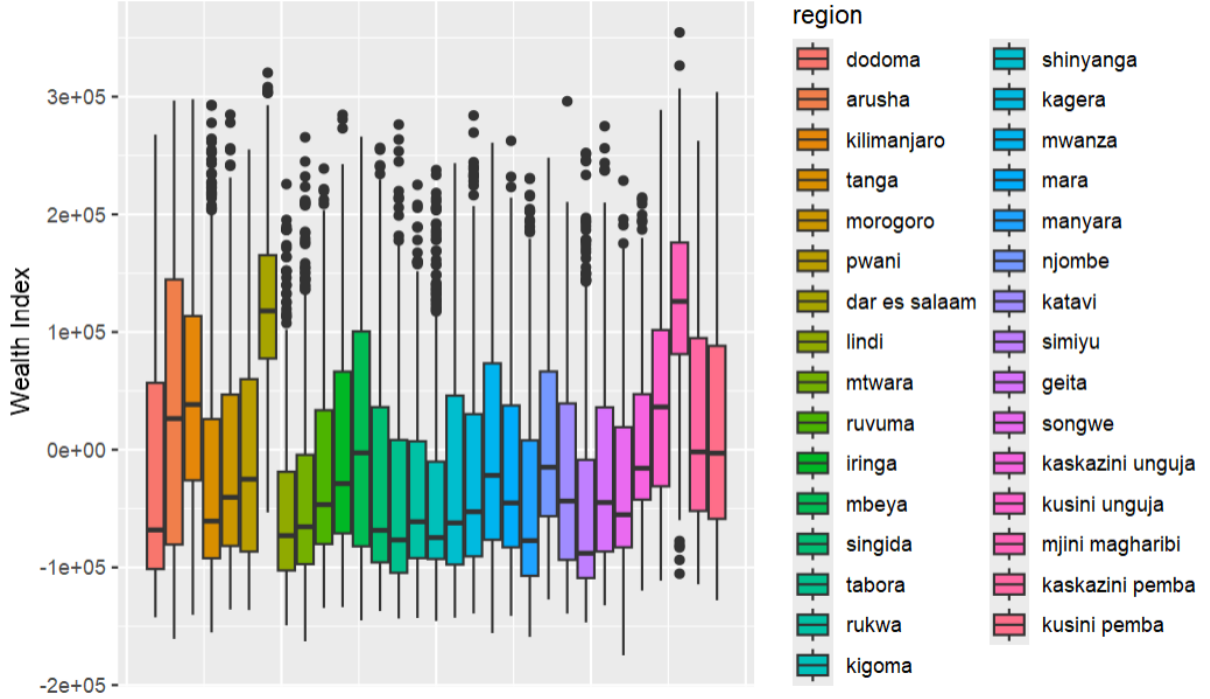


Figure A.2: Wealth by Region

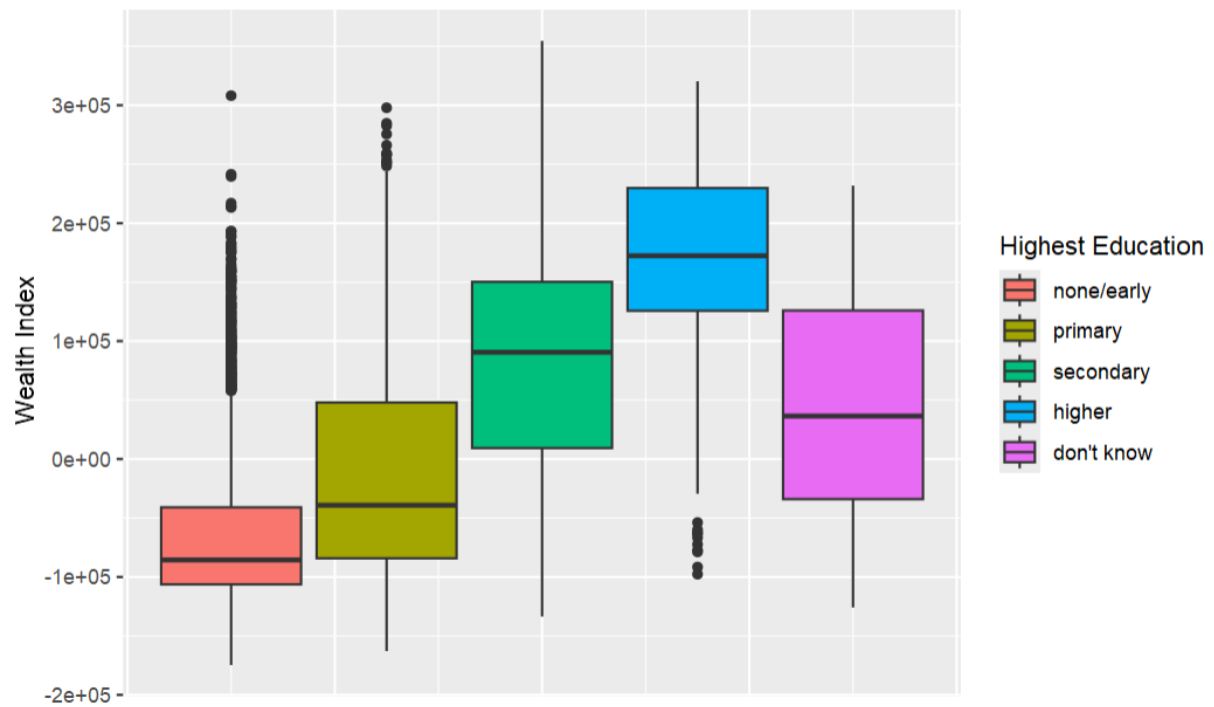


Figure A.3: Wealth by Educational Attainment

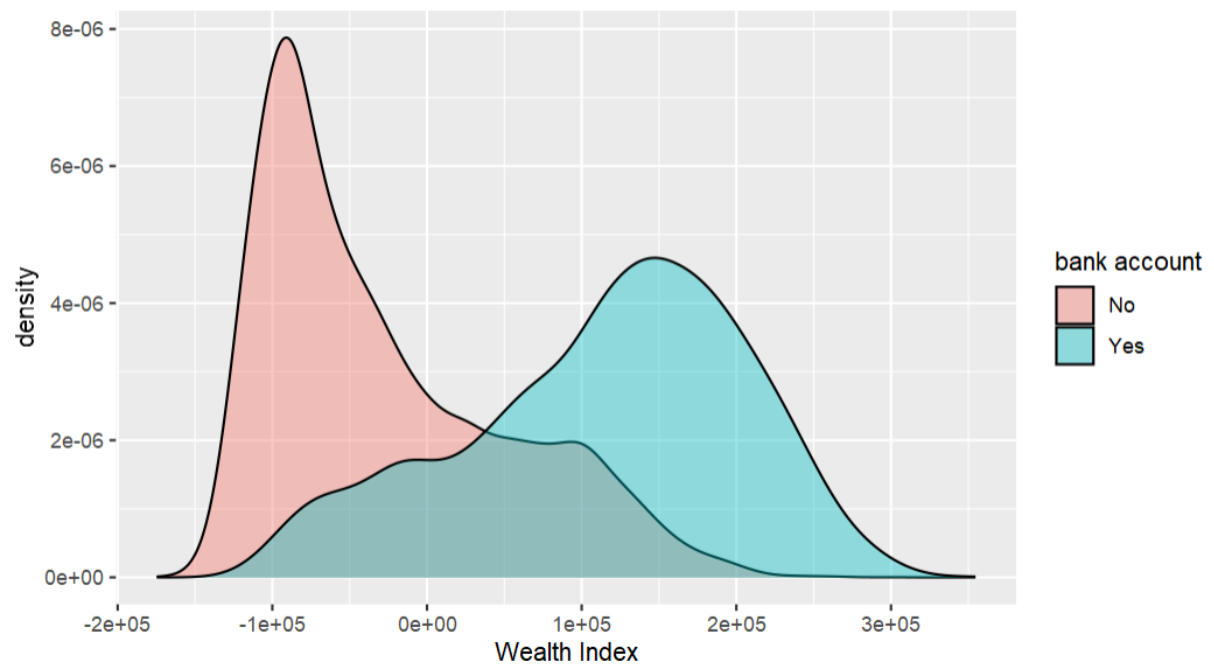


Figure A.4: Wealth by Possession of Bank Account

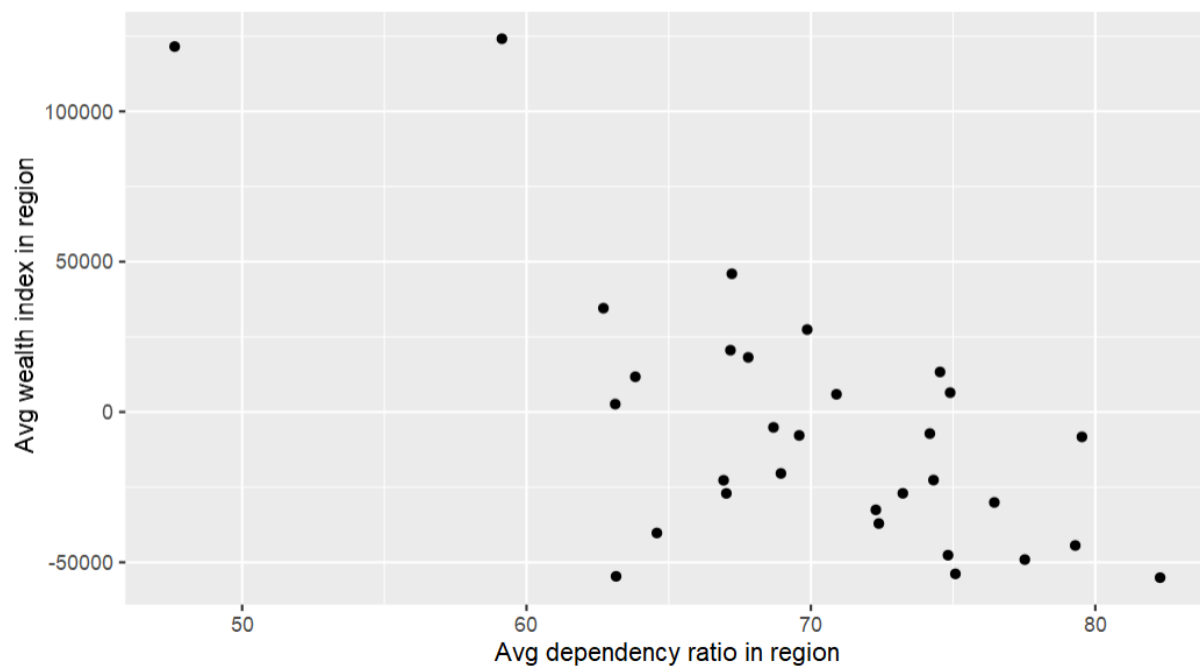


Figure A.5: Average Regional Wealth by Average Regional Dependency Ratio

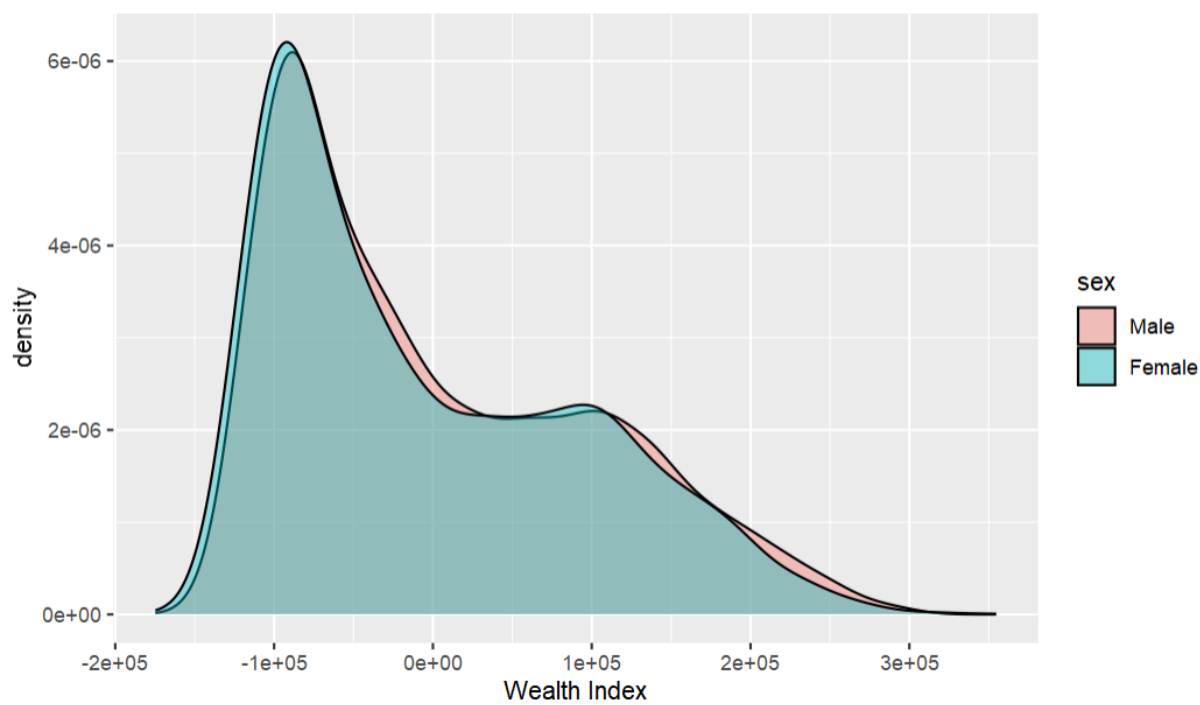


Figure A.6: Wealth by Sex of Household Head

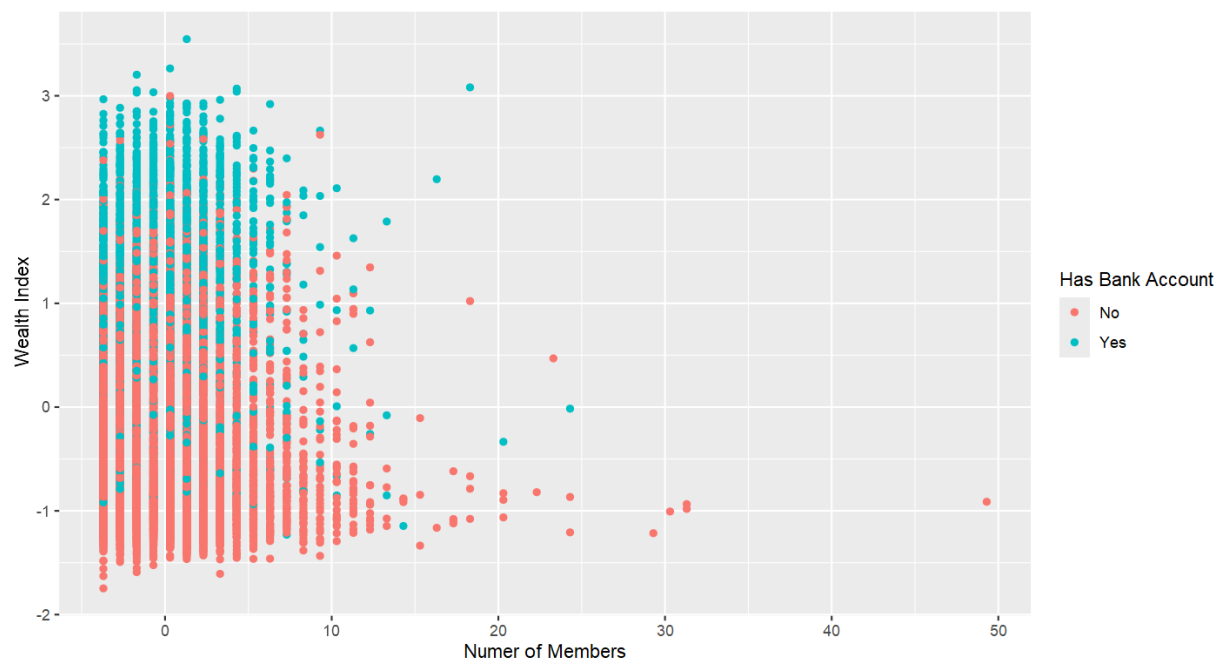


Figure A.7: Wealth by Number of Members

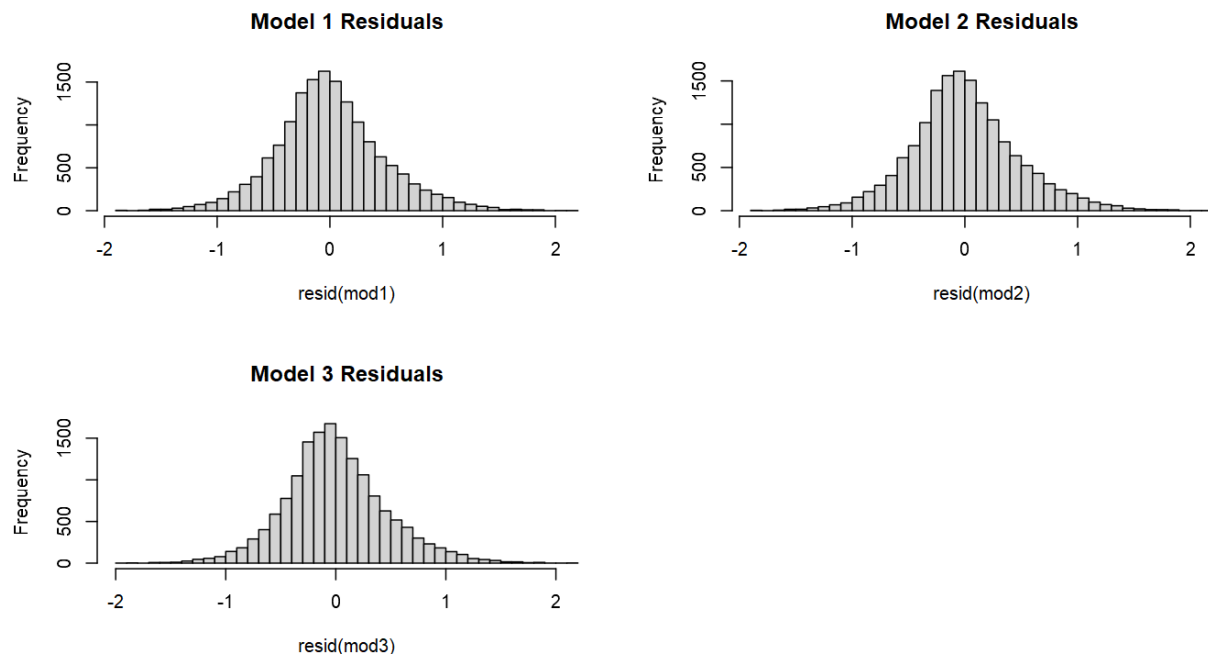


Figure A.8: Residual Histograms

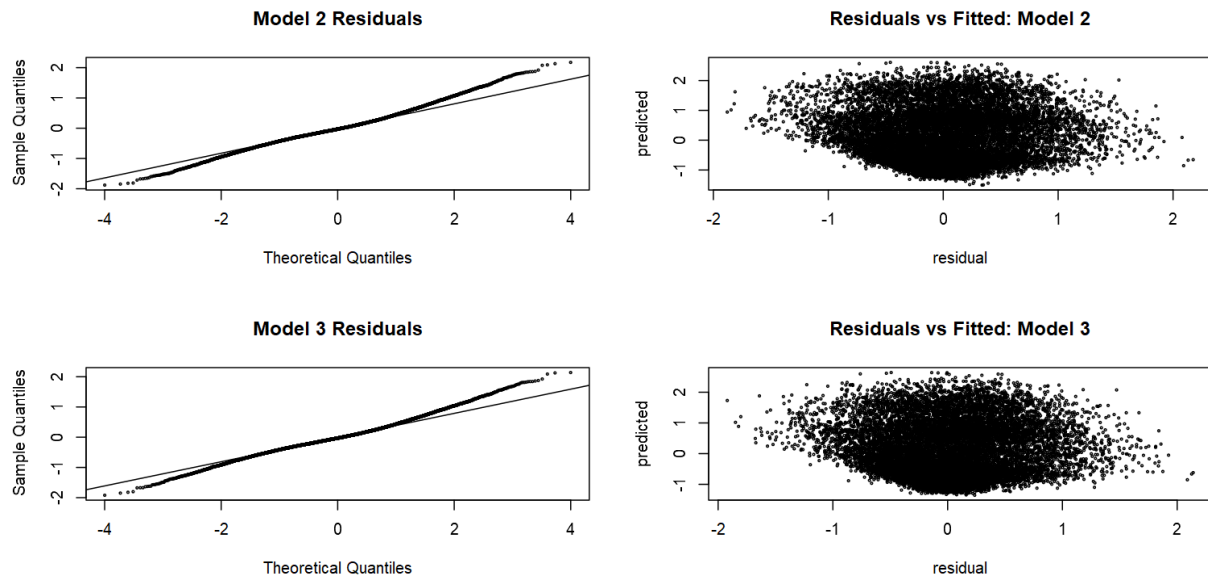


Figure A.9: Check Residual Normality and Homoscedasticity - Models 2, 3

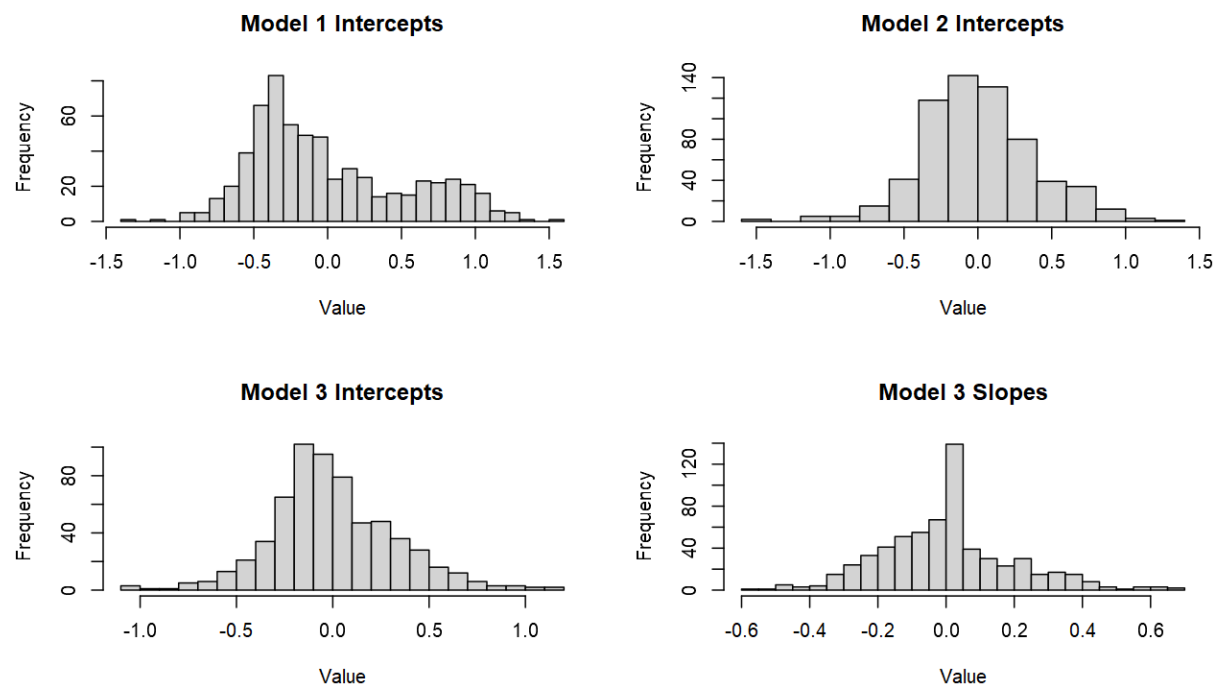


Figure A.10: Histograms of Random Effects

B Software and R Code

Data management and mathematical procedures were conducted using the `lme4` library in R. The `ggplot2` library was used to create visualizations, while other libraries were used to compute various summary statistics. A streamlined R script for model implementation is provided here. Mathematical notations and this document were compiled with L^AT_EX.

```
### Load Relevant Libraries
library(tidyverse)
library(haven)
library(lme4)
library(sjstats) # for ICC

### Data Pre-Processing: Selection, Renaming, Engineering, Transforming
tanzania_1 <- read_dta("TZHR81FL.DTA",
                      col_select = c(
                        hv001, # cluster number (PSU)
                        hv002, # household number
                        hv005, # household sample weight
                        hv009, # number of usual members
                        hv022, # sampling strata
                        hv024, # region
                        hv025, # urban/rural residence
                        hv219, # sex of household head (9:missing)
                        hv247, # has bank account (9:missing)
                        hv271, # wealth index (continuous)
                        shreg2, # analytical region
                        hv106_01) # household head education (9:missing)
) %>% mutate(hv005 = hv005 / 1e6)

# rename columns
colnames(tanzania_1) <- c("ea", "house_id", "hh_weight", "HH_SIZE",
                          "sampling_strata", "ADMIN", "RES_TYPE", "HHEAD_SEX",
                          "HASBANK", "HHWEALTH", "ANA_REG", "HHEAD_EDUC")

# fix factor names
vars <- c("HHEAD_EDUC", "RES_TYPE", "ADMIN", "ANA_REG")
for (v in vars) {
  lbls <- attr(tanzania_1[[v]], "labels")
  tanzania_1[[v]] <- factor(tanzania_1[[v]],
                           levels = lbls,
                           labels = names(lbls))
}
rm(vars, lbls, v)

# rename factor levels (improve readability)
levels(tanzania_1$HHEAD_EDUC) <- c("none/early", "primary", "secondary",
```

```

"higher", "don't know")

# calculate dependency ratio
ages <- read_dta("TZHR81FL.DTA",
               col_select =c(hv105_01: hv105_56))
tanzania_1$DEPRATIO <- apply(ages,1,
                             function(row){
                               100 * sum(row %in% c(0:14,65:99))/
                               sum(row %in% 15:64)
                             })
rm(ages)
# fix dividing by 0 issue, scale variable
tanzania_1$DEPRATIO <- pmin(tanzania_1$DEPRATIO, 100)

# mean center HH.SIZE at 0
tanzania_1$HH.SIZE <- tanzania_1$HH.SIZE -
  mean(tanzania_1$HH.SIZE)

# scale wealth in 100,000s
tanzania_1$HHWEALTH <- tanzania_1$HHWEALTH /1e5

### Model Fitting
## hv001 = ea, hv024 = region, hv025 = residence, shreg2 = govt
## For 'education': none/early is the reference
## For 'admin': dodoma is the reference
## For 'ana_reg': mainland urban is the reference
## For 'residence': urban is the reference
## For 'HHEAD.SEX': males are the reference
## —> intercepts for household with none/early
#   education male as household head, in dodoma or mainland urban or
#   urban dodoma (depending upon fixed effect used), average number of
#   members, no dependents (DEPRATIO=0)

## Model 1: FE for Administrative Region, Random Intercepts
mod1 <- lmer(HHWEALTH ~
             factor(HASBANK) + factor(HHEAD.SEX) + HH.SIZE + DEPRATIO +
             factor(HHEAD.EDUC) + factor(ADMIN) + (1 | ea),
             data = tanzania_1,
             REML = T)
summary(mod1)
performance::icc(mod1)

## Model 2: FE for Analytic Region, Random Intercepts
mod2 <- lmer(HHWEALTH ~
             factor(HASBANK) + factor(HHEAD.SEX) + HH.SIZE + DEPRATIO +
             factor(HHEAD.EDUC) + factor(ANA.REG) + (1 | ea),
             data = tanzania_1,
             REML = T)

```

```

summary(mod2)
performance::icc(mod2)

## Model 3: FE for Administrative Region, RE for bank
mod3 <- lmer(HHWEALTH ~
              factor(HASBANK) + factor(HHEAD.SEX) + HH.SIZE +
              factor(HHEAD.EDUC) + RES_TYPE + DEPRATIO +
              factor(ADMIN) + (1 + factor(HASBANK) | ea),
              # intercepts at hh, slopes for bank
              data = tanzania_1,
              REML = T)
summary(mod3)

# compute comparison statistics
AIC(mod1,mod2,mod3)
BIC(mod1,mod2,mod3)

### Assessment and Comparisons
# check normality and heteroskedasticity of residuals
par(mfrow=c(1,2))
qqnorm(resid(mod1),main="Model 1 Residuals",cex=0.3)
qqline(resid(mod1))
plot(resid(mod1),fitted(mod1),cex = 0.3,
      main="Residuals vs Fitted: Model 1",
      xlab="residual",
      ylab="predicted")

par(mfrow=c(2,2))
qqnorm(resid(mod2),main="Model 2 Residuals",cex=0.3)
qqline(resid(mod2))
plot(resid(mod2),fitted(mod2),cex = 0.3,
      main="Residuals vs Fitted: Model 2",
      xlab="residual",
      ylab="predicted")

qqnorm(resid(mod3),main="Model 3 Residuals",cex=0.3)
qqline(resid(mod3))
plot(resid(mod3),fitted(mod3),cex = 0.3,
      main="Residuals vs Fitted: Model 3",
      xlab="residual",
      ylab="predicted")

par(mfrow=c(2,2))
hist(resid(mod1),breaks=30,main="Model 1 Residuals")
hist(resid(mod2),breaks=30,main="Model 2 Residuals")
hist(resid(mod3),breaks=30,main="Model 3 Residuals")

```

```

# check normality of random effects
par(mfrow=c(2,2))
qqnorm(ranef(mod1)$ea[,1], cex=0.3, main='Model 1 Intercepts ')
qqline(ranef(mod1)$ea[,1])

qqnorm(ranef(mod2)$ea[,1], cex=0.3, main='Model 2 Intercepts ')
qqline(ranef(mod2)$ea[,1])

qqnorm(ranef(mod3)$ea[,1], cex=0.3, main='Model 3 Intercepts ')
qqline(ranef(mod3)$ea[,1])

qqnorm(ranef(mod3)$ea[,2], cex=0.3, main='Model 3 Slopes ')
qqline(ranef(mod3)$ea[,2])

par(mfrow=c(2,2))
hist(ranef(mod1)$ea[,1], breaks=30, main="Model 1 Intercepts",
      xlab="Value") # bi-modal: urban/rural split
hist(ranef(mod2)$ea[,1], breaks=20, main="Model 2 Intercepts",
      xlab="Value")
hist(ranef(mod3)$ea[,1], breaks=20, main="Model 3 Slopes",
      xlab="Value")
hist(ranef(mod3)$ea[,2], breaks=20, main="Model 3 Slopes",
      xlab="Value")
# most near 0, very small difference in bank effect across EA

```

[All additional codes and visualizations can be found in our GitHub Repository](#)